Baldivis
Secondary College

Year 10 - 2016

Investigation 4

Semester Two Term 4

Linear Equations, Parallel and Perpendicular lines

This investigation will be completed in class with help and guidance provided for the initial question. You will then complete a second question to fully test your understanding of linear relationships. No guidance or hints will be provided for this. **You may have one A4 note page and a calculator.**

AIM: *In this assessment you will be investigating the triangle formed from three linear equations and calculate its area in square units. The formula to find the area of a triangle is:*

Area = ½ x base x perpendicular height

Mark:	/22
	%

Name: Marking Key Class: ____________

1. Calculate the area of the triangle formed (in square units) from the following three linear equations:

$y = x + 4$ $\qquad$ $y = 3x$ $\qquad$ $y = -x + 4$

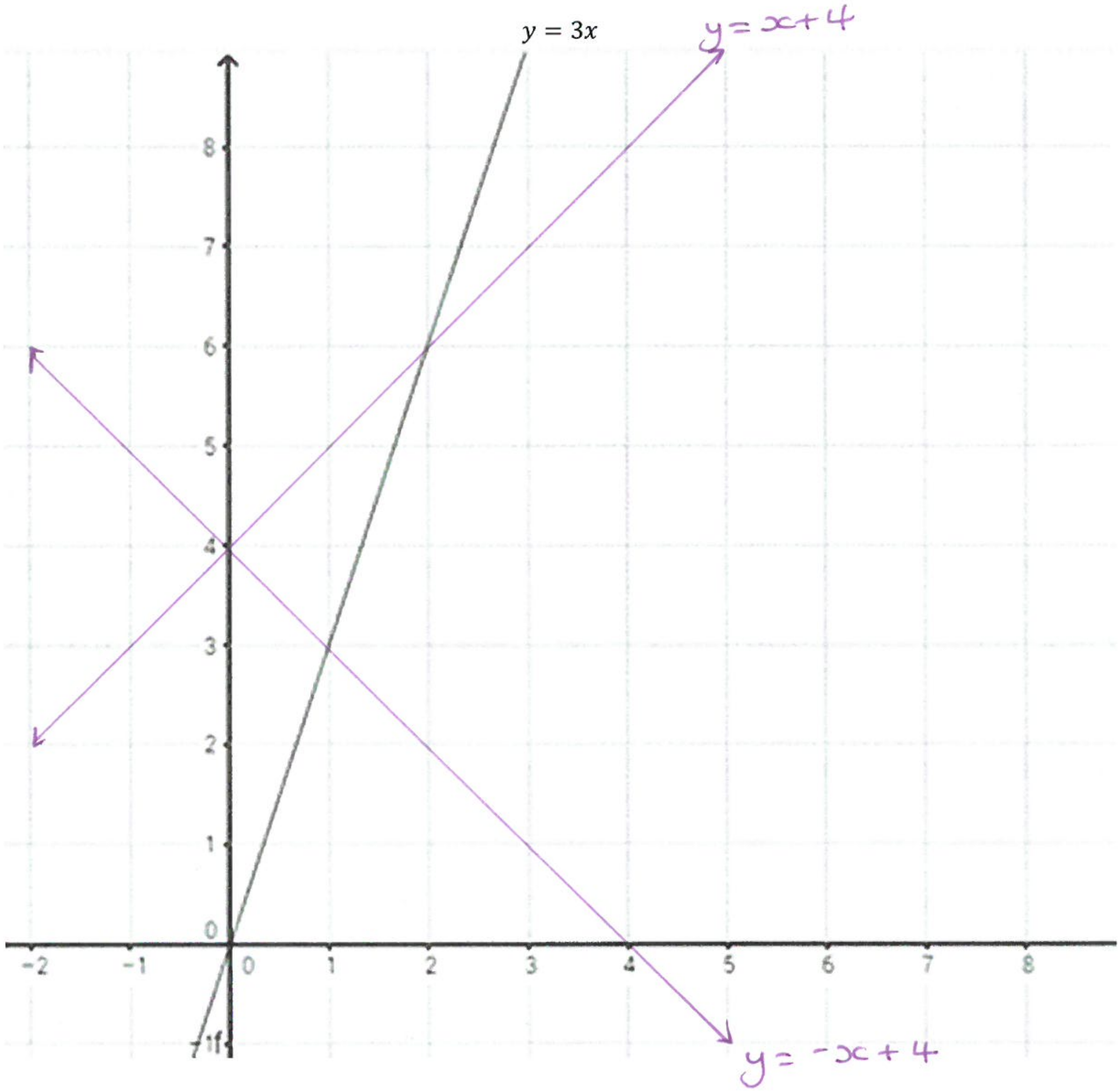
First, find the gradient and y intercept of all three lines:

Equation	Gradient = $\frac{rise}{run}$	Y intercept
$y = x + 4$	1	4
$y = 3x$	3	0
$y = -x + 4$	-1	4

Two of the lines must be perpendicular.

$y = x + 4$ and $-x + 4$ are perpendicular because their gradients are the negative reciprocal of each other. These lines will cross at right-angles. The gradients of perpendicular lines multiply to make -1. $m_1 \times m_2 = -1$

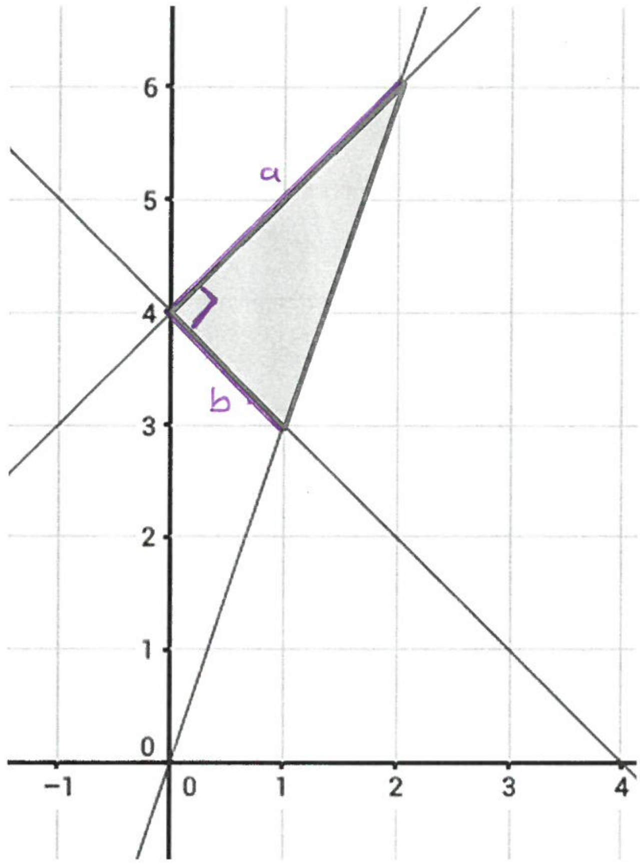
Next; graph the lines on the axes provided



What are the coordinates of the points where the lines cross?

(0,4) (1 , 3) and (2 , 6)

Identify the triangle, the right angle and the sides you need to measure.



In this case, the lines we need to measure run from:

(0,4) to (2,6) and (0,4) to (1,3)

a For the first line from (0,4) to (2,6):

Length = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Length = $\sqrt{(2-0)^2 + (6-4)^2}$

Length = $\sqrt{(2)^2 + (2)^2}$

Length = $\sqrt{4+4}$

Length = $\sqrt{8}$

Length = 2.83 square units (to 2 d.p)

b For the second line from (0,4) to (1,3)

Length = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Length = $\sqrt{(1-0)^2 + (3-4)^2}$

Length = $\sqrt{(1)^2 + (-1)^2}$

Length = $\sqrt{1+1}$

Length = $\sqrt{2}$

Length = 1.41 square units (to 2 d.p)

The area of the triangle is therefore:

Area = ½ x 1.41 x 2.83 = 2.00 square units. (2dp)

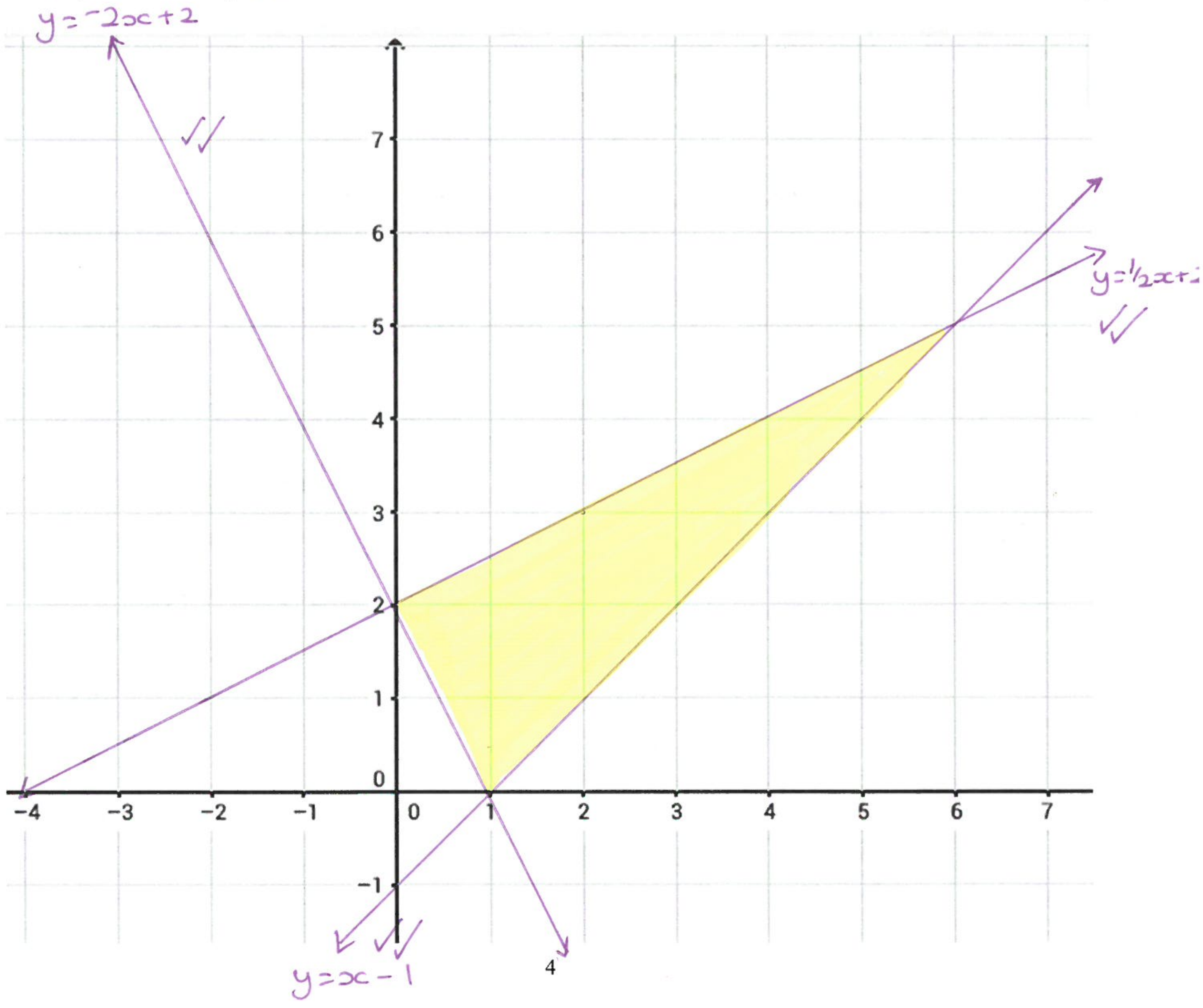
2. Calculate the area of the triangle formed from the following three linear equations:

$y = -2x + 2$ $\qquad$ $y = \frac{1}{2}x + 2$ $\qquad$ $y = x - 1$

[6]

Equation	Gradient	Y intercept
$y = -2x + 2$	-2 ✓	$+2$ ✓
$y = \frac{1}{2}x + 2$	$\frac{1}{2}$ ✓	$+2$ ✓
$y = x - 1$	1 ✓	-1 ✓

Graph the three equations [6]



Identify the triangle formed on the graph ✓ [1]

List the coordinates of the points where the lines cross to form the triangle. [3]

(0,2) ✓ (6,5) ✓ (1,0) ✓

Find the lengths of the lines which meet at right angles [4]

Hint: Length = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

(0,2) to (6,5)

$= \sqrt{(6-0)^2 + (5-2)^2}$ ✓

$= \sqrt{36 + 9}$

$= \sqrt{45}$

$= 6.71$ (2dp). ✓

(0,2) to (1,0)

$= \sqrt{(1-0)^2 + (0-2)^2}$

$= \sqrt{1^2 + (-2)^2}$ ✓

$= \sqrt{1 + 4}$

$= \sqrt{5}$

$= 2.24$ (2dp) ✓

What is the area of the triangle formed in square units? [2]

Area = ½ x base x perpendicular height

Area $= \frac{1}{2} \times 2.24 \times 6.71$ ✓

$= 7.52$ square units (2dp) ✓

END OF INVESTIGATION